7 Human–Machine Relation

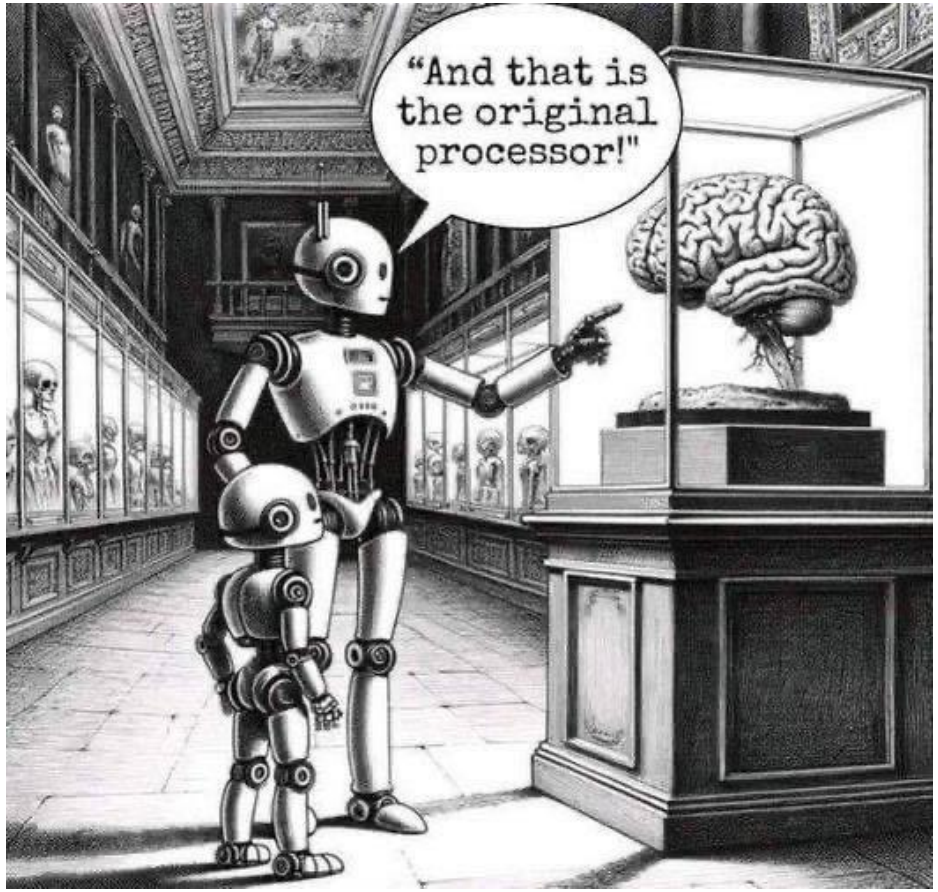
GEB2508 AI and Society

Review of Last Class

- Empire, Colonialism, and Orientalism
  - Scientific Racism
- Science and Technology as Colonial Infrastructures
  - S&T as Tools of Empire
  - Empire as A Laboratory for S&T
- Postcolonial Perspectives of AI
  - Data Colonialism
  - Cultural Hegemony
  - Global Labor Regime & Supply Chain

Content

- Structure and Agency
  - Technological Neutrality Thesis
  - Agency and Responsibility
- Human-Machine Relation
  - Cyborg and Posthumanism
- Socio-Technical System
  - Technology Diffusion and Substitution
  - Path Dependence & Lock-in
  - AI as Socio-Technical Infrastructure



Structure

- **Patterned** social relations that become separate from the life-world and exert their own **power** (Habermas)
- The framework of society, the pattern of social relationships, the **network** of social institutions, and the **organization** of social activities (Durkheim)
- The totality of regularized patterns of social action (Weber)

Agency

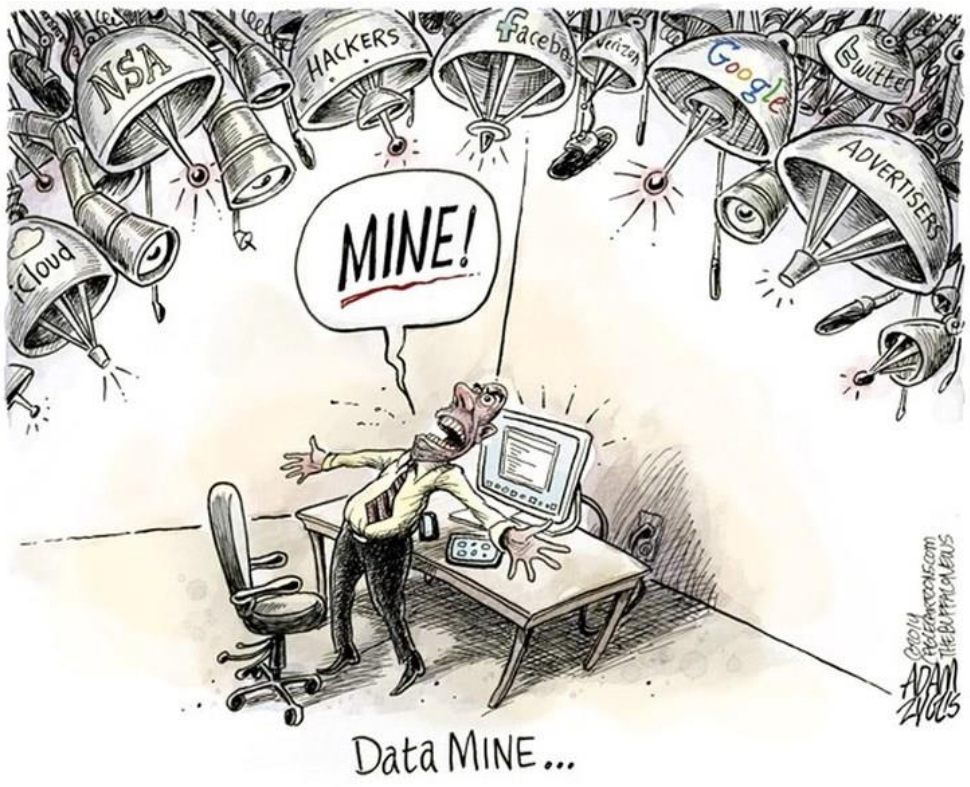
- Agency: one's ability to make a difference in the world
  - Agents: entities having agency
  - Minimal vs. strong agency → routine vs. transformative action
- Autonomy and free will
  - Degrees of autonomy: freedom from vs. freedom to
  - **Unequal** resources shape agency: agency as socially **distributed**

Structure vs. Agency

- Do social structures determine human behavior, or does human agency determine society?
  - The **objectivist** view: individuals' behavior is largely shaped by social structures through processes of socialization
  - The **subjectivist** view: individuals can exercise free will; social structures are **products** of individual actions rather than their causes.
- Duality of structure: social structures exist through practices

Animate and Inanimate Agency

- Agency or intentionality of non-human actors and **inanimate** objects
  - Infrastructure; technology; etc.
- ANT (Actor-Network Theory): agency is **distributed** across a network of human and non-human actors
  - Opening the black box: examining **stabilized** technology whose inner workings become invisible once taken for granted

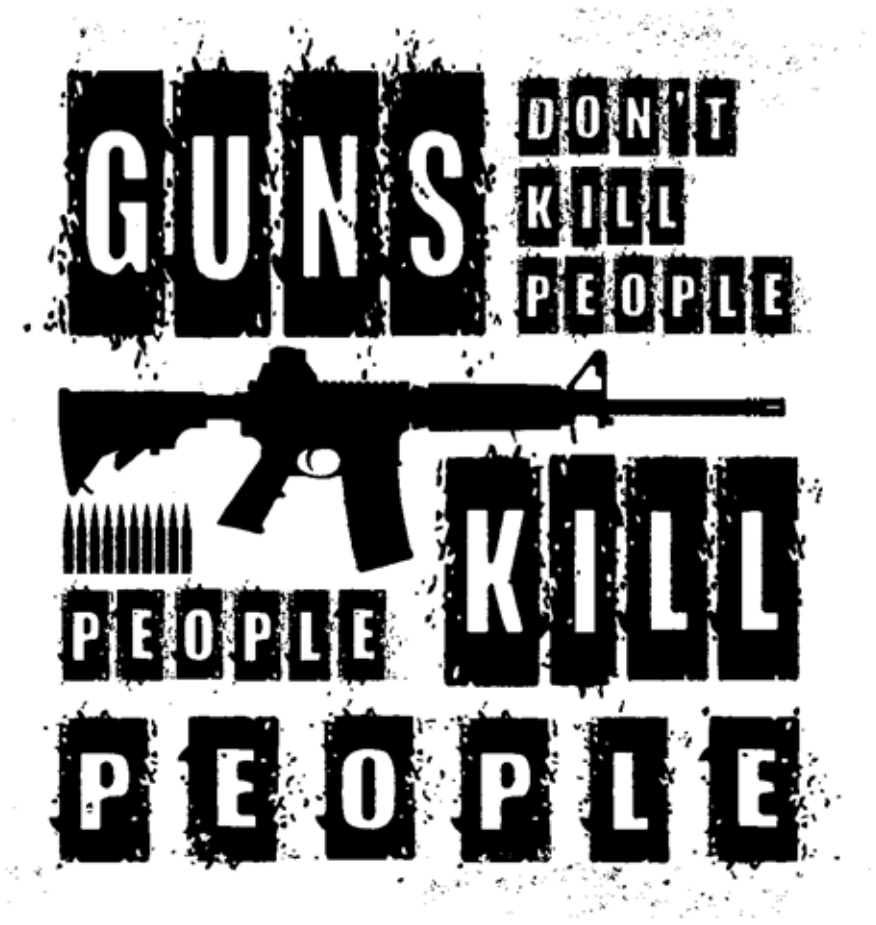


Is Technology Value Neutral?

- Structure vs. agency → technology and value
- The **Value Neutrality Thesis** (VNT): technology has no inherent moral value; only through how people use it
- Counter-argument: designers embed values and assumptions into technology → shift accountability from **end-users** to **designers**

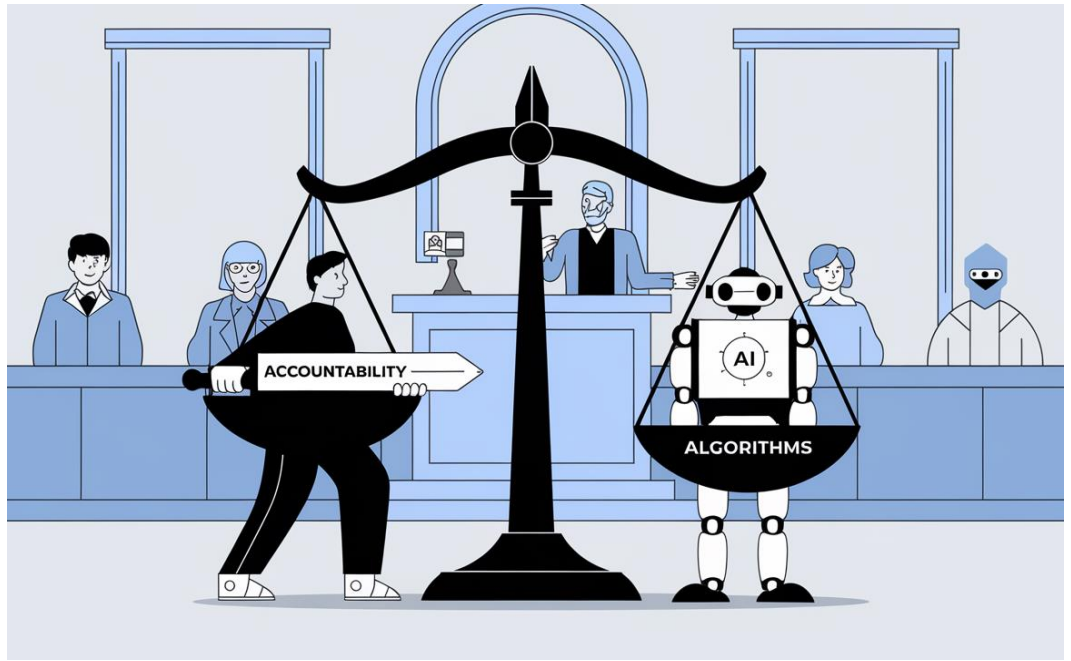
The Case of Gun Control

- "Guns don’t kill people, people kill people" goes the popular slogan of the National Rifle Association (NRA) in America.
- The availability and ease of access to firearms significantly contribute to the lethality and frequency of acts of violence.
- Do we shape our technology, or does technology shape us?



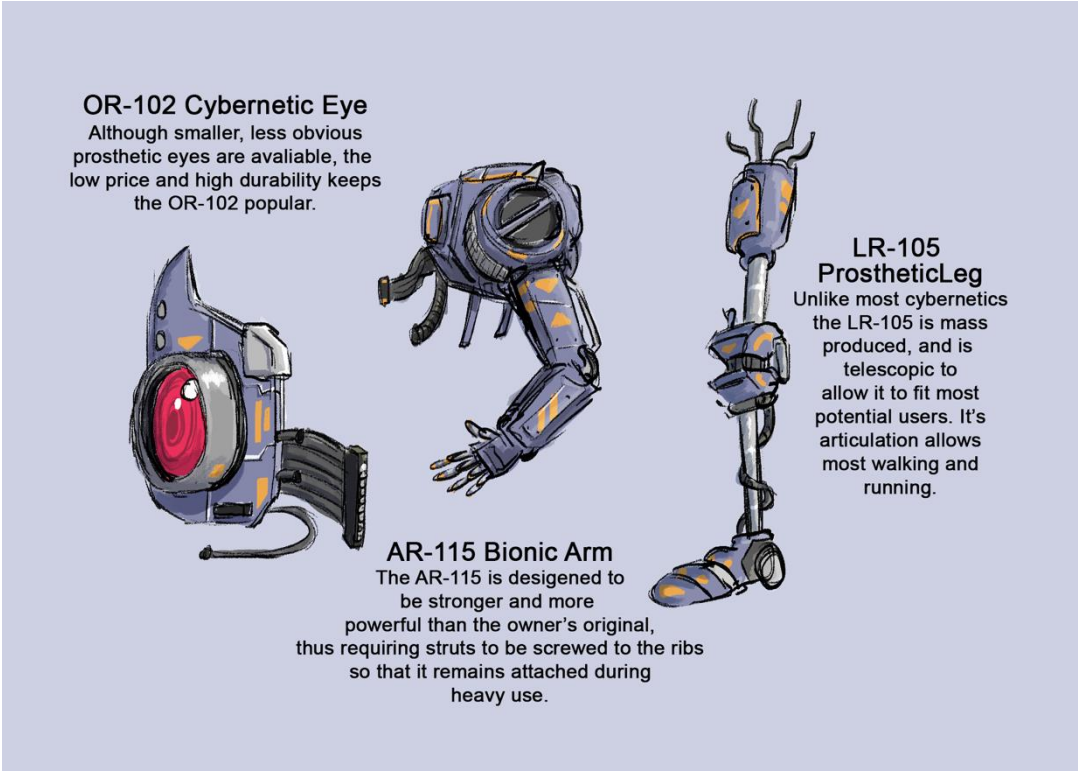
AI's Agency and Accountability

- **Distributed** agency in AI systems
  - Humans + machines + institutions
  - No single agent causes harm.
- Accountability gaps: who is responsible when AI acts?
  - Blame shifting
  - Tool vs. teammate vs. decision-maker
- **Rights** vs. **liability**
  - AI legal personhood?

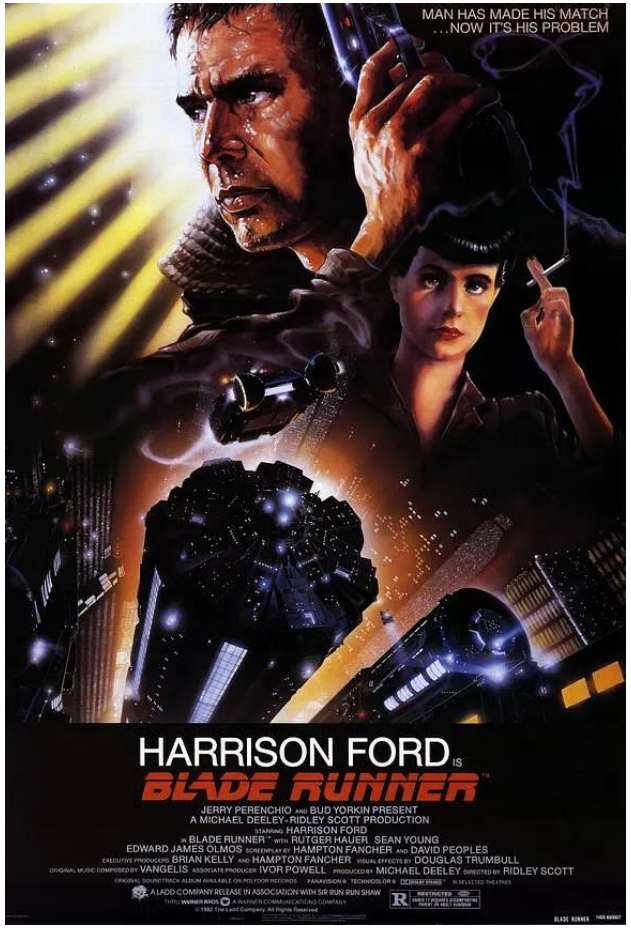


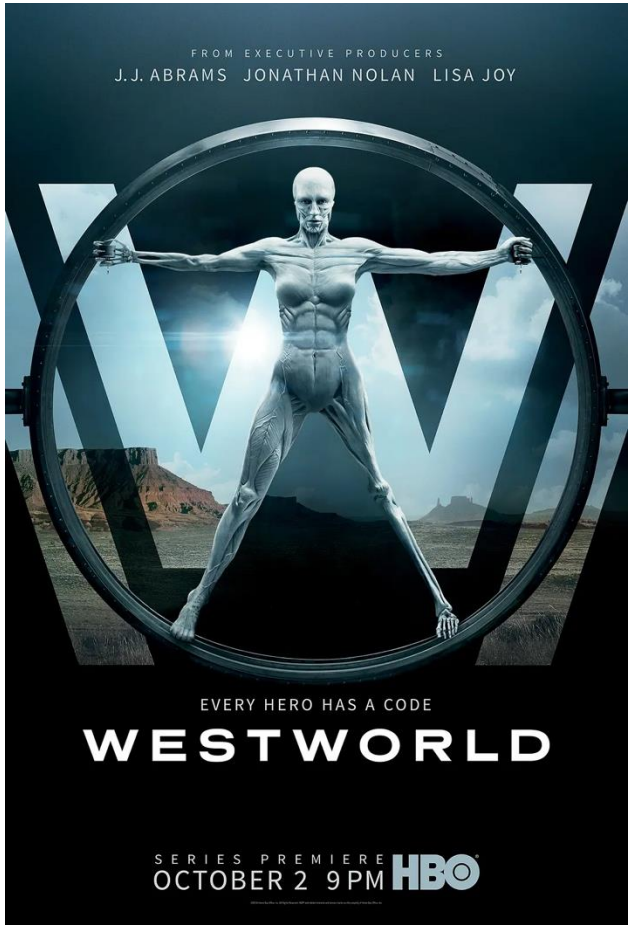
Cyborg

- Cyborg: cybernetic organism
  - Cyborgs in everyday life: pacemakers, prosthetics, cochlear implants, smartphones, wearables, etc.
- Critique of pure, natural, or essential human identity
  - **Boundary breakdowns**: human/machine, natural/artificial
- Cyborg as a relational assemblage



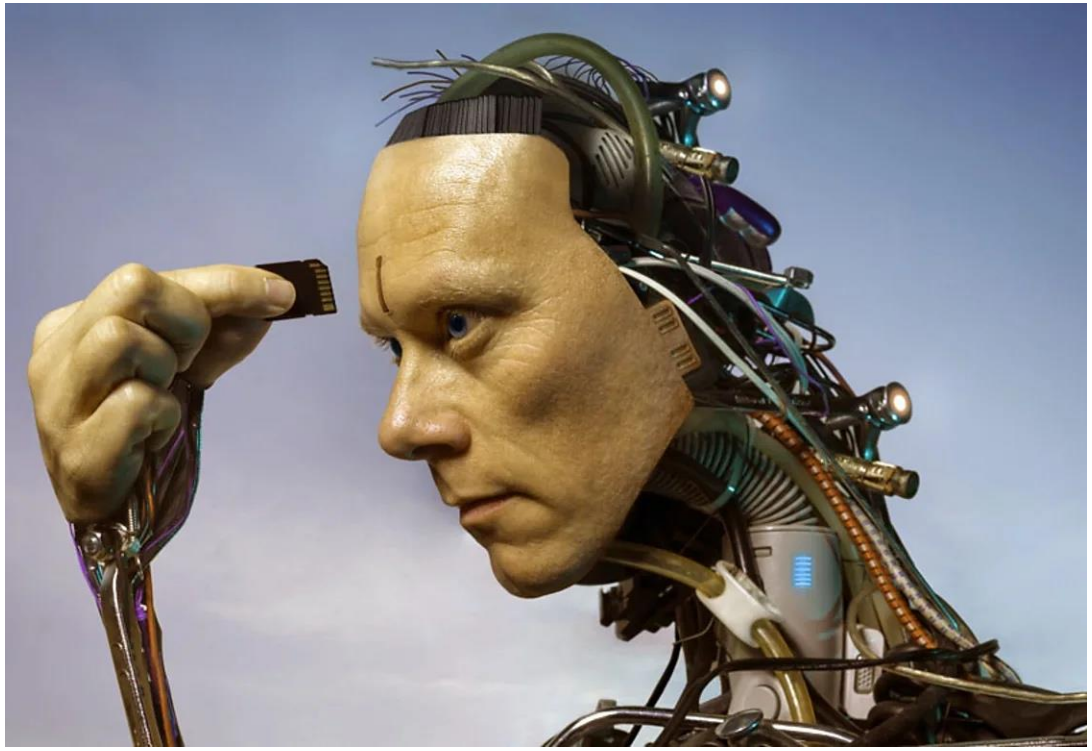
Cyborg in Popular Culture





Cyborg’s Implication for AI and Robotics

- How technology **reorganizes** human agency and cognition
- AI as cognitive extension
  - E.g., algorithmic trading, diagnostic systems
- Robotics as physical extension
  - E.g., drones, exoskeletons, prosthetics
- Human-machine symbiosis and co-production
  - Brain–computer interface (BCI)

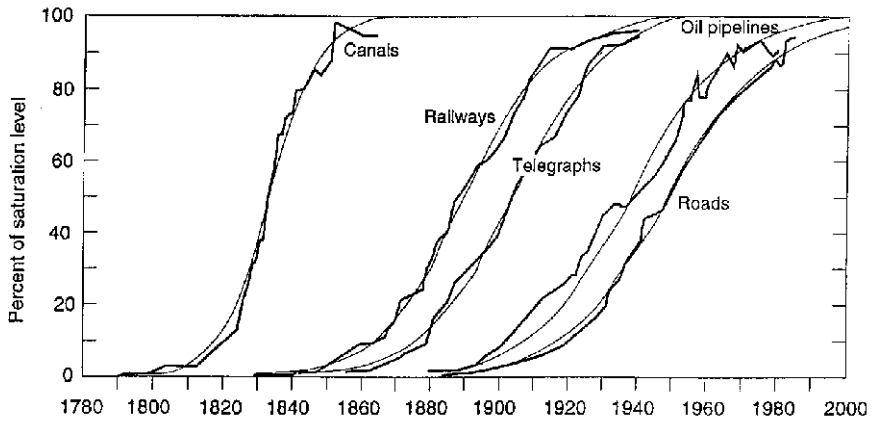


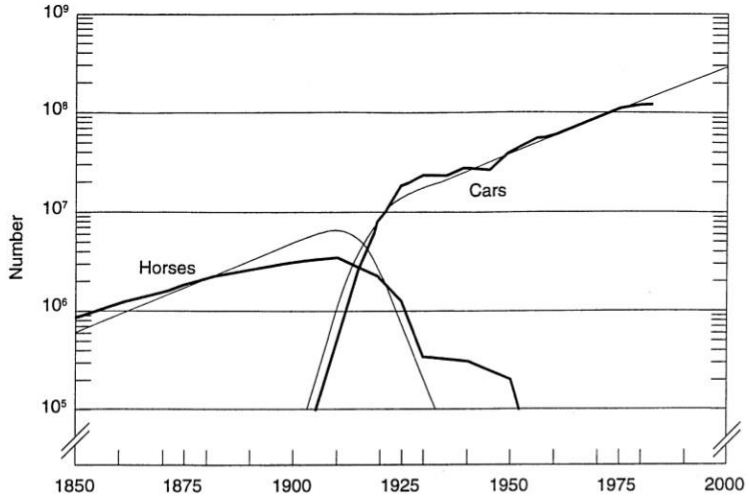
Posthumanism and AI

- From “how machines become human” to examining the **assumptions** about human
  - Intelligence, creativity, agency
- **Decentering** the human as the sole reference point of agency
- The “disembodied mind” fantasy
  - Material and infrastructural AI
  - Hidden labor and bodies

Technology Diffusion and Substitution

- Differential adoption rates: technologies diffuse **unevenly** across populations (e.g., by socio-economic status, age, geography)
- Technological substitution: new technology replaces an incumbent one
  - Horse-drawn carriages → petrol automobiles
  - Film photography → digital photography
  - Incandescent bulbs → LED lighting



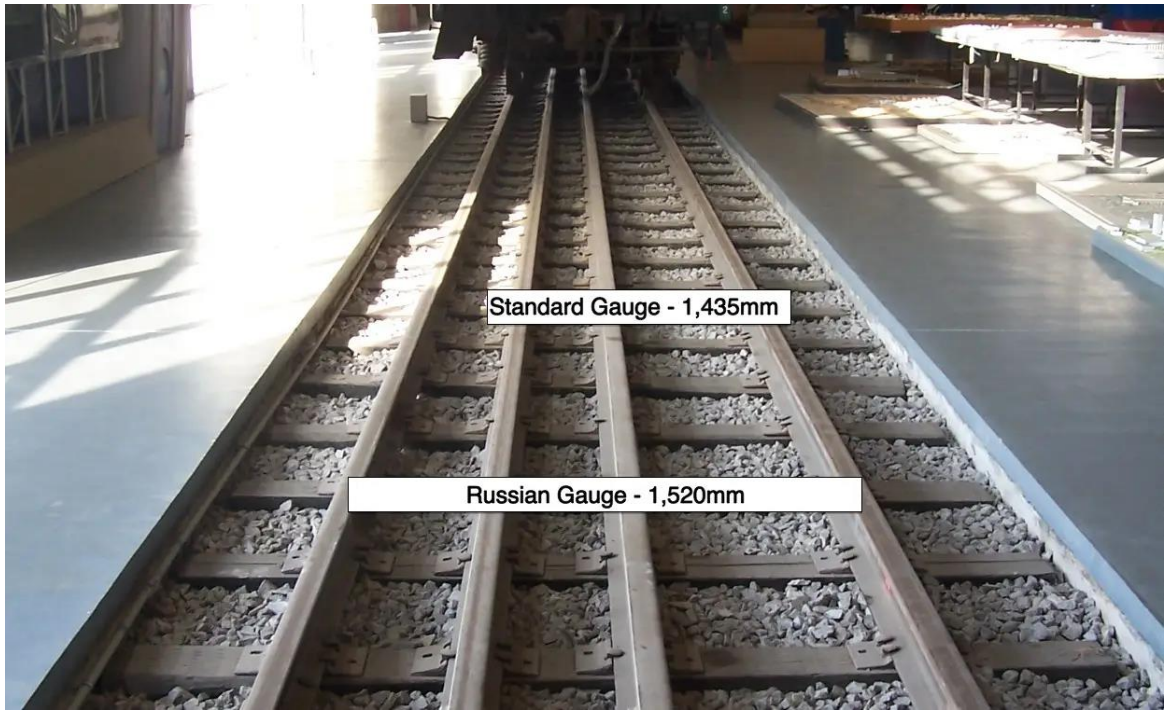


Path Dependency

- History matters: new technologies do not automatically replace old ones
- Path dependence refers to how current outcomes and choices are **contingent on** and **conditioned by** past events.
- People stick with standards not because they are **optimal**, but because they became **entrenched**.

Technological Lock-in

- Path dependence can lead to technology lock-in: society stuck with an established technological standard even when it is no longer optimal
- Increasing returns: the more a technology is adopted, the more advantages it accumulates → **self-reinforcing** trajectory

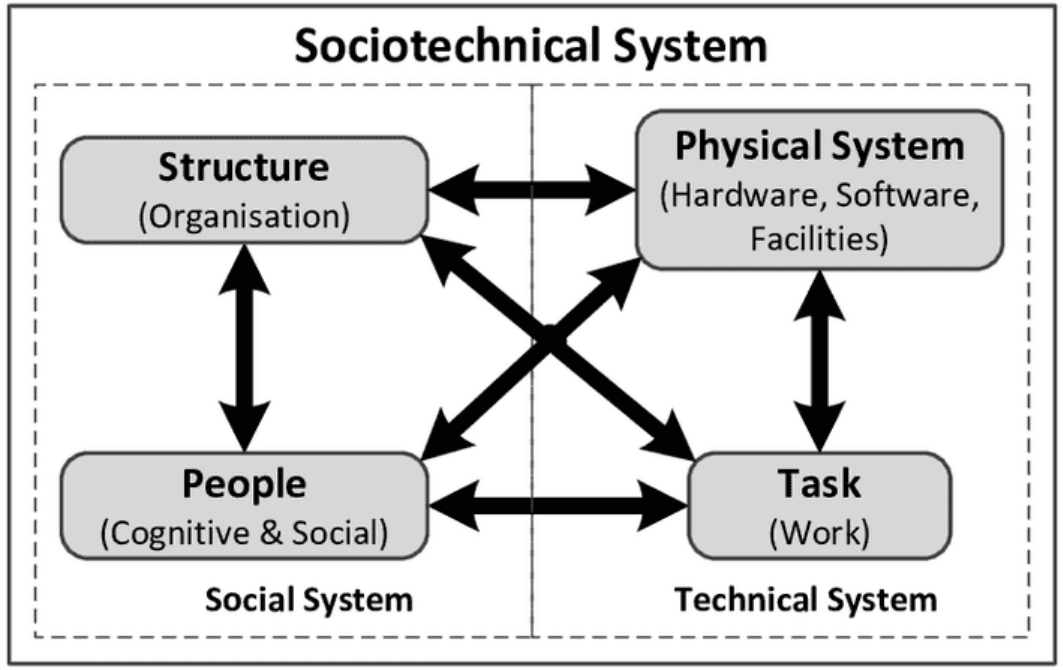


The Case of the QWERTY Keyboard

- The concept of path dependence first gained attention through scholars' interpretation of the persistence of the QWERTY standard.
- "We teach kids how to use QWERTY because it's everywhere. Why is QWERTY everywhere? Because we teach kids to use it."

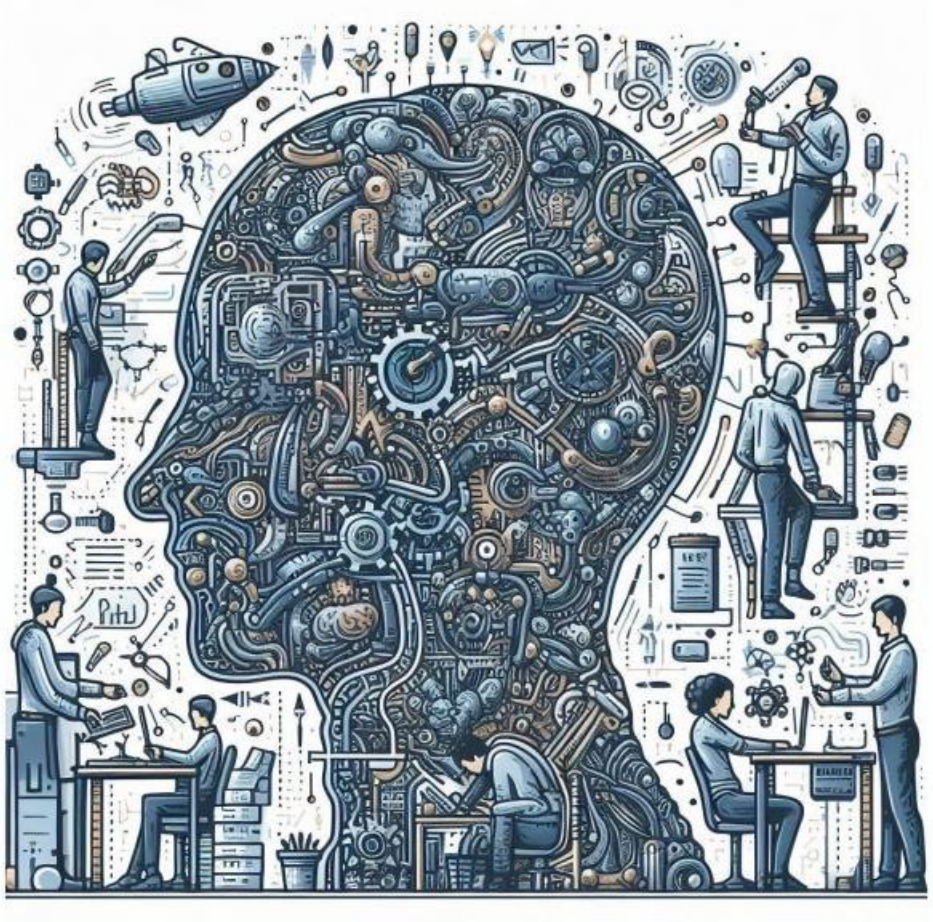
Socio-Technical Systems

- **Co-constitution** of:
  - Technical systems (hardware, machines, infrastructures)
  - Social systems (people, rules, norms, institutions)
- Technologies are inseparable from social relations, values, and power.
  - **Centralized** power vs. diffused responsibility



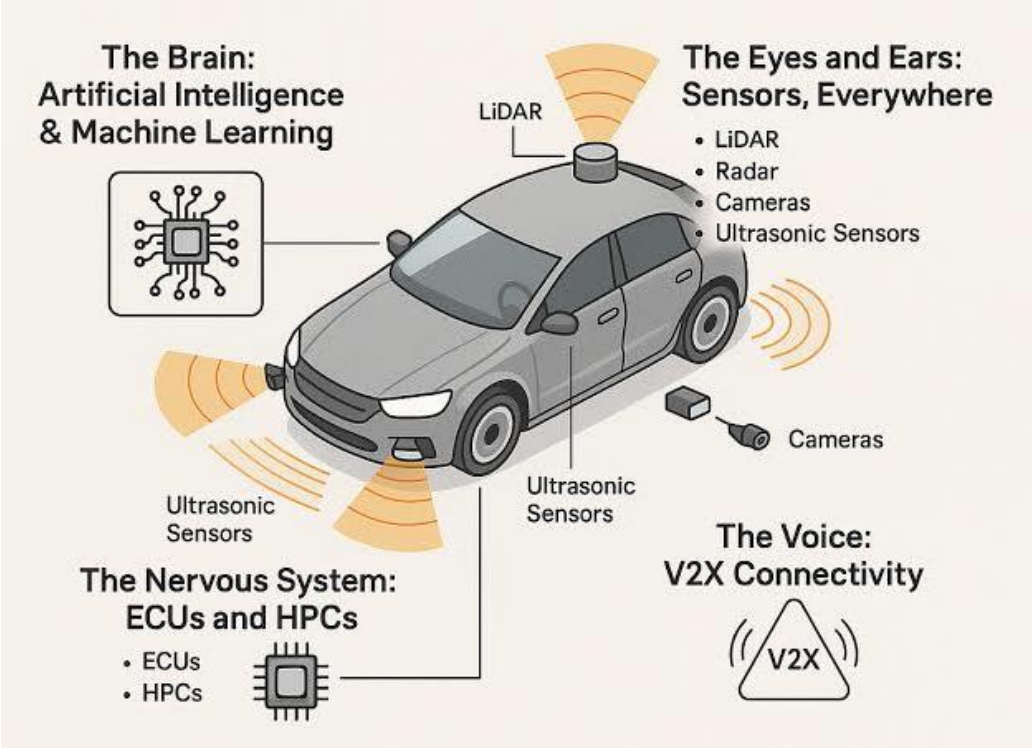
AI as Socio-Technical System

- AI ≠ algorithm alone
  - Data are social products
  - Modeling choices are value-laden
  - Deployments are embedded in organizations
  - Feedback loops: interaction, adaptation, and path dependence
- Misplaced agency and displaced responsibility
  - “The algorithm decided”



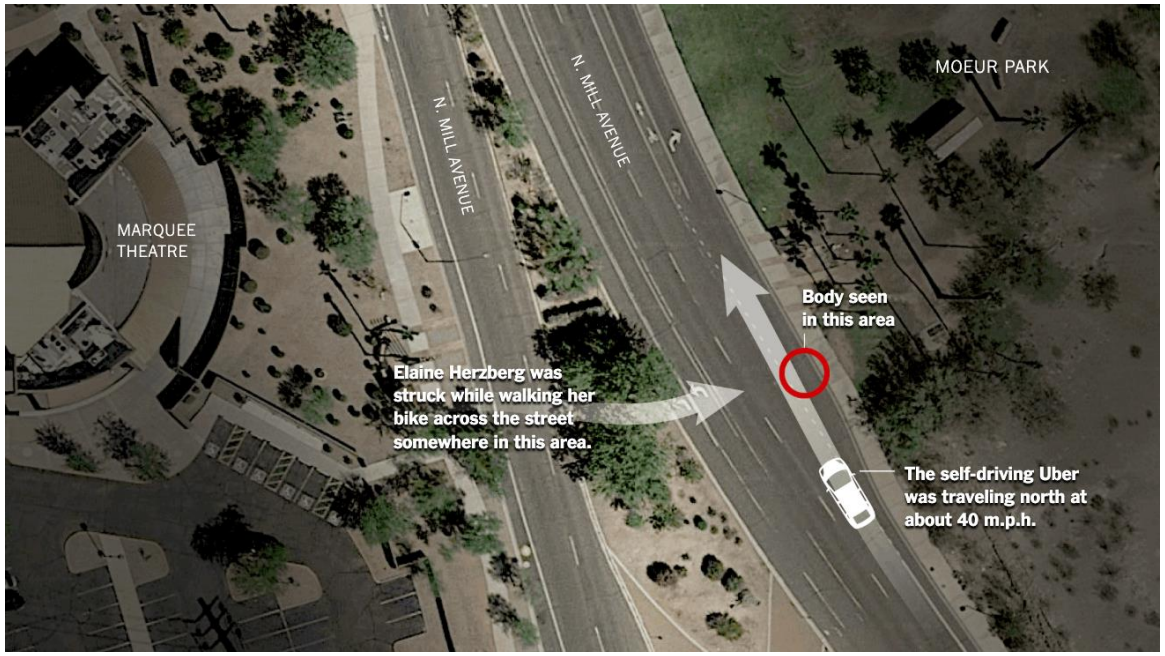
The Case of Self-Driving Cars

- Assemblage and distributed intelligence
- Car accidents as the outcome of **layered** decisions
  - Accountability must be traced across the socio-technical network that **produced the conditions**
- **Human-centered** liability and the accountability gap



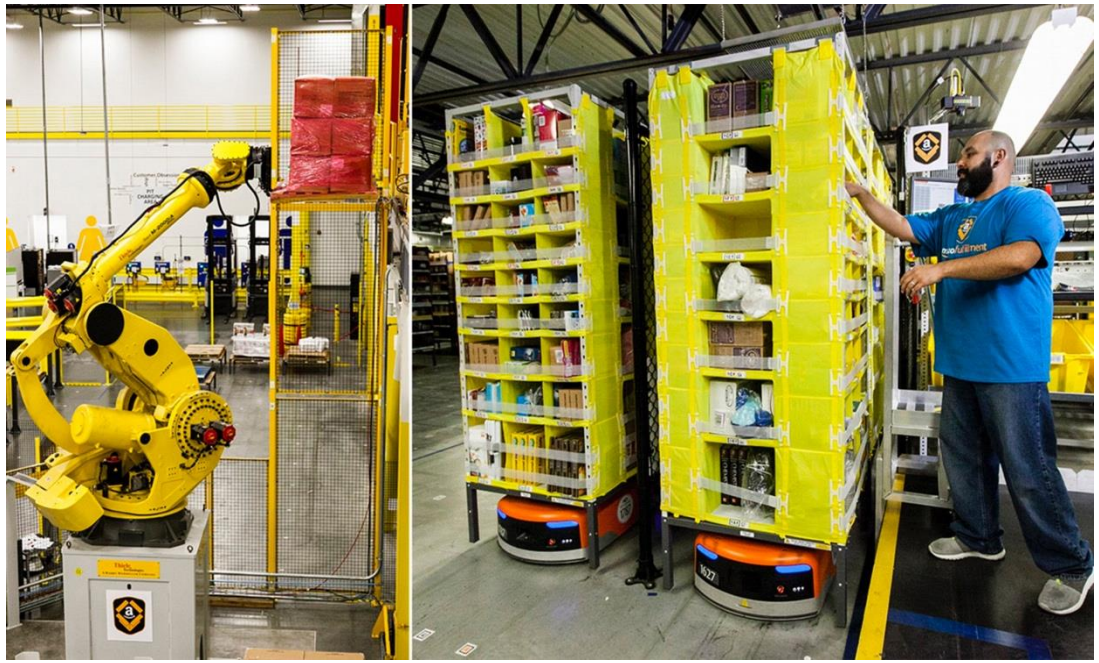
The 2018 Uber Autonomous Vehicle Fatality

- Uber test AV in self-driving mode struck and killed pedestrian in Arizona
- Accountability: responsibility individualized onto safety driver; no criminal charges against Uber
- Agency was distributed, but liability was concentrated on the **least powerful** actor.



Socio-Technical Transition

- Socio-technical transitions: fundamental and long-term shifts in socio-technical **regimes** (stable configuration of technologies and social arrangements & practices)
  - Horse-drawn transport → automobile
  - Electrification
  - Digital revolution
- Inherently **political** process

AI-Driven Work Reorganization

- AI reshapes labor and organizations
- Manufacturing: AI alters production planning and how workers respond to disruptions
- Agriculture: precision farming shifts decisions toward data-driven management
- Healthcare: AI reconfigures clinical workflows